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TERM III

Date: 13/06/2025

EXAM:/100

CLASS: SENIOR 6

SUBJECT: COMPUTER SCIENCE

Combinations:

MATHEMATICS-PHYSICS-COMPUTER SCIENCE (MPC)

MATHEMATICS-COMPUTER SCIENCE-ECONOMICS(MCE)

Instructions

Attempt all questions. **(All questions are compulsory).**

For open-ended questions, write the questions number on provided sheet.

For multiple-choice questions, write only the letter of the correct answer.

1. Define: /4
 - a) Web server b) Thread
 - c) computer graphics d) Multimedia
2. Which statement best explains the role of cache memory in a computer system? /1
 - A. It stores programs permanently C. It temporarily holds frequently accessed data
 - B. It helps display graphical output D. It increases storage space
3. A friend wants to connect a digital camera to their computer. Which port would you recommend? /1
 - A. VGA C. USB
 - B. Serial D. PS/2
4. Which hardware upgrade best improves video editing performance? /1
 - A. Replace LCD with CRT C. Upgrade 256 MB RAM to 4 GB
 - B. Add external speakers D. Install a 56k modem
5. Why is preventive maintenance important in a computer lab? /1
 - A. Slows computer speed C. Extends hardware lifespan
 - B. Prevents software updates D. Enables faster browsing
6. A computer plays no sound, but the media player runs. What should you check first? /1
 - A. Restart the media player C. Check speakers and mute settings
 - B. Increase the screen brightness D. Reinstall the operating system
7. How many host addresses are available in a Class C network? /2
 - A) 254 B) 256 C) 65,536 D) 16,777,214
8. Discuss 2 benefits and 2 challenges of artificial intelligence in modern computing. /2
9. You are tasked with advising a school on setting up a new computer lab. Suggest four key preventive maintenance practices and justify why each is important. /2
10. What is the main function of a UPS? /1
 - A. Boost internet speed C. Provide power during outages
 - B. Store backup files D. Cool the compute
11. Which of the following is NOT a correct ergonomic practice? /1
 - A. Using adjustable chairs
 - B. Placing monitor at eye level
 - C. Using poorly lit environments
 - D. Keeping wrists straight while typing
12. Swapped Out Processes Are Managed By: /1
 - A. Process Control block C. Medium-Term Scheduler
 - B. Long-Term Scheduler D. Short-Term Scheduler
13. Which license type allows users to freely use, modify, and share software? /1
 - A. Proprietary License C. Freeware License
 - B. Shareware License D. Open-Source License
14. What factor should be considered before purchasing software? /1
 - A. Color of the packaging C. Compatibility with hardware
 - B. Developer's nationality D. Screen resolution

15. Which of the following is NOT a universal gate? /1
- | | |
|---------|-------------------------------|
| A. NAND | C. AND |
| B. NOR | D. Both A and B are universal |
16. What is the decimal equivalent of the binary number 1010? /1
- | | |
|-------|------|
| A. 12 | C. 9 |
| B. 10 | D. 8 |
17. Which operation is used in a sum of products expression? /1
- | | |
|--|---------------------|
| A. Multiplication followed by addition | C. Only addition |
| B. Addition followed by multiplication | D. Only subtraction |
18. Which Boolean identity simplifies $A + A \cdot B$? /1
- | | |
|-------|------------|
| A) A | C) $A + B$ |
| B) AB | D) B |
19. Explain the difference between 1's complement and 2's complement with examples for converting -13 into binary. /3
20. What type of control structure checks a condition before executing a block of code? /1
- | | |
|----------|-------------------|
| A. FOR | C. REPEAT...UNTIL |
| B. WHILE | D. GOTO |
21. Which of the following is true about a one-dimensional array? /1
- | | |
|---|-----------------------------------|
| A. It stores values in rows and columns | C. It contains only string values |
| B. It stores values in a single list | D. It is used for decision-making |
22. What is the output of an algorithm that converts 10 from base 10 to binary? /1
- | | |
|---------|---------|
| A. 1100 | C. 1110 |
| B. 1010 | D. 1001 |
23. In a REPEAT...UNTIL loop, when is the condition evaluated? /1
- | | |
|------------------------------|-------------------------|
| A. Before the loop starts | C. After each iteration |
| B. At the middle of the loop | D. Only once |
24. What is the purpose of using arrays in control structures? /1
- | | |
|--|------------------------------|
| A. To declare functions | C. To assign constant values |
| B. To store a group of related variables | D. To create flowcharts |
25. Design an algorithm and flowchart that reads the grades of 10 students, calculates the average grade, and outputs whether each student passed or failed based on a pass mark of 50. /4
26. To reference storage of a variable in main memory, two operators, namely size of and address of (&) may be used. Using sample code, differentiate between the two operators. /3
27. A linear list of elements in which deletion can be done from one end (front) and insertion from the other (rear) is known as-----/2
28. Using clear example explain inheritance in java. /4

- 29. Which access specifier is used to hide data from outside the class? /1**
- A. public
 - B. protected
 - C. private
 - D. global
- 30. What is function overloading in C++? /1**
- A. Redefining base class methods
 - B. Same function name, different parameters
 - C. Using multiple classes
 - D. Same variable name in different functions
- 31. Which function is used to allocate memory to an object in C++? /1**
- A. malloc()
 - B. alloc()
 - C. new
 - D. create
- 32. a) What are benefits of Java Collection Framework? /2**
- b) Write a java program that performs the following operations using a linked list: /6
- i) Create a linked list and assign it to a variable named district.
 - ii) Add the following districts to the list: Gakenke, Rubavu, Gasabo, Nyagatare, Nyabihu.
 - iii) Add Bugesera district to the front of the list.
 - iv) Insert Karongi district at the 4th position (i.e., index 3).
 - v) Replace the Gasabo district with Nyarugenge.
 - vi) Retrieve and display the second district in the list (index 1).
 - vii) Display the size of the list.
 - viii) Remove the last district from the list.
- 33. Which keyword is used to make a function a friend of a class? /1**
- A. friend
 - B. private
 - C. protected
 - D. virtual
- 34. What happens in External Fragmentation? /1**
- A) Memory Blocks Are Wasted Inside the Files
 - B) Free Memory Is Broken into Small Pieces
 - C) Files Become Unreadable
 - D) File Size Increases
- 35. Which Of the Following Is Not A Core Component Of Apache Tomcat? /1**
- A. Jasper
 - B. Catalina
 - C. Coyote
 - D. Weblogic
- 36. What Is the First Step in A Servlet's Life Cycle? /1**
- A. Destroy Method Is Called
 - B. Service Method Is Called
 - C. Servlet Instance Is Created
 - D. Servlet Class Is Loaded
- 37. What Does the Status Code 500 Indicate? /1**
- A. Success
 - B. Client Error
 - C. Server Error
 - D. Not Modified
- 38. a) List 6 Properties of File. /3**
- b) Give 4 Examples of Networking Devices. /2
- 39. Which property ensures that data is delivered to the correct recipient? /1**
- A. Latency
 - B. Security
 - C. Reliability
 - D. Bandwidth

40. a) The ability to change the logical schema without altering external schemas is called...../2
 b) The constraint used to link records between two tables is a...../2
41. Write a Java program to demonstrate multilevel inheritance. Create three classes: Animal, Dog, and BabyDog, where Dog inherits from Animal and BabyDog inherits from Dog. Each class should have a method that prints a specific message (eat, bark, and weep respectively). Then, in a Test class, create an object of BabyDog and call all three methods. /5
42. Authorization is the process of: /1
- A. Encrypting files
 - B. Identifying users
 - C. Granting access rights
 - D. Removing user accounts
43. Explain the role of each of the following protocols:/3
- A. DHCP (Dynamic Host Configuration Protocol)
 - B. ARP (Address Resolution Protocol)
 - C. ICMP (Internet Control Message Protocol)
44. Choose the suitable word(s) to fill the following blanks: /5
- a. An instance of a program execution is called _____ (Process, Instruction, Procedure, Function)
 - b. A scheduler which selects processes from secondary storage device is called_____ (Short term scheduler, Long term scheduler, Medium term scheduler, Process scheduler).
 - c. The _____ is used as the repository for any information of the process. (Process state, deadlock, CPU, PCB).
 - d. _____is the module that gives control of the CPU to the process selected by the short-term scheduler. (Dispatcher, processor, aging).
 - e. _____ Scheduler select the process that is in the ready queue to execute and allocate the CPU to it. (Short-term scheduler, Long term scheduler, Medium term scheduler, dispatcher).
45. Assume that a database is connected to a Visual Basic form using the ADODC control. Provide the methods or actions required to: /5
- a) Add a new record b) Move to the next record c) Move to the previous record,
 - d) Move to the first record e) Move to the last record f) Delete the current record.

46. Provide html and CSS code of the below form:/5

APE RUGUNGA STUDENT FORM

This form is mandatory to be filled by new student of APE RUGUNGA.

Full Name:

Address:

Date:

Time:

In O'level subjects to sit for entrance exams are :

- a. Biology
- b. Chemistry
- c. Mathematics
 - Algebra
 - Statistics
 - Probability
- d. Physics
- e. English for all

In A' level Choose the combination :

Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

- a) Draw the Gantt chart using the First Come First Serve (FCFS) scheduling algorithm. /2
- b) Calculate the Waiting Time (WT) and Turnaround Time (TAT) for each process. /2
- c) Compute the average Waiting Time and average Turnaround Time. /1

End!